

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2195

West Rutland 100 kV
February 21, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2195. The Interconnection Customer proposes a 183.8 MW solar plus storage facility interconnecting at the West Rutland 100 kV substation in the Ohio area. The request was received on October 02, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 617 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,101,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
West Rutland / Sage Road Substation 100 kV	W Rutland / Weiderstein 100 kV Line	107.2	166	100.0
Sage Road / Albion 100 kV	W Rutland / Weiderstein 100 kV	105.9	162	100.0
Calapooya / Albion 100 kV	W Utland / Weiderstein 100 kV	104.5	144	100.0
Weiderstein / Killen 100 kV	W Rutland / Sage Rd 100 kV	100.4	177	100.0
Calapooya / Bryant 100 kV	Toledo Bend / Killen 500 kV	100.3	146	26.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W Rutland / Sage Road 100 kV line	Reconductor line	\$2,418,000	100% (\$2,418,000)
Albion / Sage Road 100 kV	Reconductor line	\$2,161,000	100% (\$2,161,000)
Calapooya / Albion 100 kV Line	Reconductor line	\$1,906,000	100% (\$1,906,000)
Killen / Weiderstein Substation 100 kV line	Reconductor line	\$451,000	100% (\$451,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Calapooya / Bryant 100 kV line	Reconductor line	\$1,165,000	100% (\$1,165,000)
	Total allocated to project		\$8,101,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.